

anything. Indeed, unlike Mary, who can use her other experiences as a scaffold for understanding color—much like Helen Keller using associations from senses like touch and smell to construct a color scheme despite being blind and deaf (“*Pink makes me think of a baby’s cheek, or a gentle southern breeze*”)—LLMs have none of this kind of sensory scaffolding.

Essential aspects of human cognition—emotions, intuition, and other subjective experiences—so far remain incomputable and therefore cannot be fully replicated by computational algorithms. Even if everything about human cognition were theoretically computable, actually capturing the full depth and complexity of human thought processes would likely exceed computational feasibility. The interplay of perception, memory, emotions, and social contexts creates patterns so intricate that complete replication would require exponentially increasing computational resources. In other words, creating AI systems that genuinely replicate human cognition—systems behaving like humans under all circumstances—may be fundamentally intractable from a computational perspective (Van Rooij et al., 2024). AI systems—constrained by their current algorithms, data, and embedded assumptions—face inherent challenges in replicating human cognition. This limitation parallels Gödel’s famous discovery in mathematics: Any formal system sophisticated enough for basic arithmetic will contain true statements that cannot be proven within that system (Fokas, 2023). Similarly, AI systems face intrinsic constraints in capturing all aspects of human thought within their computational frameworks.

To mitigate these issues, consider the following.

1. **Model selection and settings:** Different model architectures may capture different aspects of linguistic behavior (Dettki et al., 2025). Choose models based on their specific capabilities for the research question rather than assuming larger or more advanced models automatically approximate human cognition better (Zan et al., 2025; Zhou et al., 2024). Compare base models with fine-tuned versions to understand how optimization objectives affect outputs relative to human responses (Binz & Schulz, 2023a; Yax et al., 2024). Evaluate the impact of model parameters such as temperature (Li & Qi, 2025).
2. **Prompt design:** Test model sensitivity to prompt variations to distinguish between robust patterns and artifacts of specific phrasings (Brucks & Toubia, 2025). This helps identify when models are exhibiting systematic “reasoning” versus merely responding to surface-level cues in ways humans would not (Ivanova, 2025).
3. **Interpretations and applications:** Validate LLM outputs against human data, particularly when studying cognitive processes, to establish appropriate boundaries for generalizing from LLM experiments to human psychology. Make explicit that LLM performance, however impressive, stems from statistical prediction rather than human-like understanding (Shiffrin & Mitchell, 2023).
4. **Ethics:** Acknowledge model limitations in research reporting, particularly the inherent distinction between token prediction and human cognition, to prevent misinterpretation and reduce the risk of anthropomorphizing LLMs in scientific literature (Ibrahim & Cheng, 2025).

Box 2 | Biological versus Machine Optimization

Natural selection introduces a different optimization mechanism from machine training, which directly impacts how we should interpret intelligence in both systems. Biological pressures for survival and reproduction embed implicit assumptions and constraints within neural systems—termed ecological inductive biases (Richards et al., 2019). These biases enable humans to learn efficiently from physical and social interactions, manage limited cognitive resources by prioritizing selective inputs and tasks, and adapt to uncertainty

using shortcuts and flexibility. Such evolved constraints stand in contrast to the engineered bias of language models, which arise from their objective function (e.g., next-token prediction, reward maximization), design (e.g., embeddings, transformer architecture), and training data composition. Other computational models similarly have their distinctive inductive biases; for example, convolutional neural networks (CNNs) learn patterns by focusing on local spatial relationships, as their convolutional filters operate over small image regions.

A crucial distinction is that biological evolution carries historical baggage that constrains optimization in ways that shape cognitive capabilities. When environments suddenly change, biological systems cannot rebuild from scratch but must adapt using existing neural machinery, often creating suboptimal yet functional solutions. This evolutionary path dependency is further constrained by the DNA bottleneck, limiting intergenerational information transfer: All adaptive changes must be encoded in the genome, which can only store a fraction of an organism's lifetime learning. In contrast, AI systems face no such constraints. They can be redesigned from the ground up with each iteration, and their "knowledge" can be transferred directly via perfect copying, distillation into smaller models, or selective merging of capabilities—all without the genetic limitations that restrict biological information transfer.

These disparities between language models and humans in their optimization pressures manifest in both internal architectures and external behaviors (Doerig et al., 2023). Key architectural differences include:

- **Computation mechanisms:** Models rely on digital computation and continuous activation functions, contrasting with the analog computation and all-or-none neuronal firing characteristic of the human brain;
- **Information flow:** Models predominantly use feed-forward processing, whereas the human brain exhibits extensive feedback modulation across interconnected regions; and
- **Learning algorithms:** Models depend on backpropagation—an algorithm that adjusts network weights based on output errors—for learning, which is biologically implausible due to its reliance on explicit error signals, symmetric weight updates during forward and backward passes, global error propagation throughout the entire network, and discrete, stepwise updates after processing batches of data.

These architectural differences produce observable behavioral distinctions between LLMs and human cognition:

- **Data efficiency:** LLMs exhibit low data efficiency, as evidenced by sublinear scaling laws—while increasing data and model size leads to improved performance, achieving incremental improvements requires exponentially more resources;
- **Energy consumption:** LLMs display low energy efficiency, consuming megawatts of power compared to the ~20 watts used by the human brain;
- **Memory stability:** Models show susceptibility to catastrophic forgetting, where learning new tasks causes the model to forget previously learned tasks;
- **Learning approach:** Current LLMs lack active exploration and continual learning—they do not retain prior interactions or instructions across sessions—unlike humans, who learn cumulatively over time; and

- Input robustness: Models demonstrate fragility to minor input perturbations and noise, exemplified by the sensitivity of LLMs to prompt variations, namely, brittleness.

These differences in optimization mechanisms, architectural constraints, and resulting behaviors illustrate why token prediction, however sophisticated, produces a qualitatively different form of intelligence than human cognition—and thus LLMs cannot simply replace human participants in psychological research.

Box 3 | Technical versus Conceptual Limitations in Model Simulation of Human Behavior and Cognition

When evaluating language models as simulation tools for psychological science, distinguishing between technical and conceptual limitations clarifies both current research constraints and future prospects. Technical limitations arise from current implementation practices and can be mitigated through methodological refinements. These include: (1) training data biases—predominantly WEIRD-skewed text corpora with specific temporal cutoffs; (2) algorithmic artifacts—such as those introduced by RLHF that systematically alter response patterns; (3) architectural constraints—including limited context windows and memory capabilities; and (4) prompt sensitivity—where minor wording variations produce substantially different outputs. These limitations affect simulation fidelity and require various methodological controls, but are potentially addressable through engineering advancements within the existing paradigm.

Progress in addressing technical limitations is already underway. Multimodal LLMs (MLLMs) now process information across text, images, audio, and video. Reasoning-focused models (sometimes referred to as large reasoning models) integrate reinforcement learning with chains of thought, explicitly narrating intermediate reasoning steps. This scaffolding promotes deeper analysis and improves performance on tasks requiring structured, multi-step inference across domains like science, math, and code. As models learn from increasingly diverse digital media, they develop more sophisticated representations of the physical world, potentially capturing dynamics beyond human sensory reach (e.g., at micro and macro scales) and improving aspects of physical common sense derived from text alone.

Conceptual limitations, by contrast, reflect intrinsic discrepancies between model design and human cognition that define the boundaries of the current paradigm. These include: (1) grounding—LLMs lack embodied experience that situates cognition in physical and social contexts, operating solely through abstract symbol manipulation; (2) agency—LLMs have no intrinsic goals, desires, or motivations driving behavior, functioning instead as sophisticated prediction systems; and (3) subjectivity—LLMs lack conscious experience that introduces qualitative dimensions to human cognition, including emotional valence and phenomenal awareness. These limitations constrain the construct validity of LLMs as psychological models in ways not readily addressed through incremental engineering.

Even rich observational learning from video may not equate to the embodied knowledge humans acquire through direct, multisensory interaction. Without lived experience, models might still struggle to fully replicate crucial aspects of human intelligence, such as deeply grounded physical intuition, nuanced social common sense, and certain forms of